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Embedding entropy: a nonlinear measure of dynamical causality

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Research on concepts and computational methods of causality has a long history, and there are various concepts of causality as well as corresponding computing algorithms based on measured data. Here, by considering causes and effects from a dynamical perspective, we present a unified mathematical framework for the so-called dynamical causality (DC), which can detect causal interactions over time; notably, this framework covers Granger causality, transfer entropy, embedding causality and their conditional versions. Based on this framework, we further propose a causality criterion called embedding entropy (EE) to measure the DC between two variables. Moreover, its conditional version, conditional embedding entropy (cEE), is also derived for detecting conditional/direct causality. The significant advantages of EE and cEE are that they can be employed for solving not only nonlinear causal inference but also the non-separability problem, and they can reduce the scale bias in numerical calculation. The performance and robustness of EE and cEE were demonstrated through numerical simulations, and causal inference on various real-world datasets validated their effectiveness.

1. Introduction

Research on concepts and computational methods of causality has a long history. In ancient Greek times, Aristotle developed the theory of 'four causes' to explain change or movement: material cause, formal cause, efficient cause (or moving cause) and final cause (or purpose cause) [1]. Since then, especially in the contemporary era, both philosophers and scientists have continued to develop concepts of causality from multiple perspectives, with significant achievements including counterfactual theory [2,3] and potential outcome frameworks [4].

Several concepts of causality have also been proposed from a mathematical perspective. Taking objects in the world as variables, mathematical causality is represented as a function from a cause variable to an effect variable, which can also be regarded as a realization of Aristotle's efficient cause. There are several universal facts that should be considered when modelling such a cause–effect association, such as 'causes precede effects in time' and 'correlation is not causality'. To date, there have been two main mathematical approaches to exploring causality. One is statistical modelling, which was developed by statisticians and entails the causality between random variables, usually without prior knowledge of time. This approach holds that causality is contained in interactions between random variables and can be detected through statistical randomization or intervention. The potential outcome framework (the Neyman–Rubin causal model [5–7]) and the structural causal model (Pearl's

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causal calculus [8,9]) are the two most famous statistical approaches, and they have been proven to be equivalent [10]. The other approach is dynamical modelling, which has been developed mostly by experts in dynamical systems. This approach uses time series as data and aims to construct concepts of causality from a dynamical perspective. Granger causality (GC) [11], transfer entropy (TE) [12,13] and embedding causality (EC) [14,15] are the three most well-known algorithms for quantifying causal strength. Recent researches about the loss of efficacy of GC and EC can be found in [16,17], respectively. Several modifications for more accurate assessment of causality can be found in [18,19]. The latest reviews and numerical comparisons on these dynamical methods can be found in [20–23]. The dynamical approach has various applications in the real world, such as detecting brain connectivity [24].

The statistical approach mainly aims to identify edges on a directed acyclic graph by defining causality with the weakest constraints [10]. Theoretically, there exist non-distinguishable situations that belong to the same Markov equivalence class under this approach [25]. Designing statistical experiments to reveal causality is the main purpose of this approach, as opposed to computationally measuring the causal strength. By contrast, in the dynamical approach, numerical algorithms are designed to computationally measure the strength of influence of one part of a dynamical system on another. This approach can be conveniently used to analyse time-series data recorded in the real world, which are difficult to analyse through intervention experiments. In particular, the computational results of the dynamical approach can provide causal relations between measured variables, which can be theoretically ensured from a dynamical viewpoint. However, GC mainly considers linear causality. TE extends GC to nonlinear cases, but it cannot solve the non-separability problem [15,26]. EC uses the delay embedding technique to overcome the non-separability problem, but it uses local linear prediction with the Pearson correlation coefficient (PCC) as the index. Besides, the theoretical connection between these approaches and concepts remains unclear from a dynamical perspective.

In this paper, we focus on giving a mathematical definition of causality under the dynamical approach and proposing a dynamical causal inference algorithm. The theory aims to overcome the problems of nonlinearity, non-separability and scale bias. Specifically, in the following sections, we first propose a dynamical causality (DC) framework based on a general dynamical equation. The DC is defined based on whether the present effect variable relies on the historic trajectory of the cause variable. DC emphasizes the causal interactions induced by dynamics over time. Then, GC, TE and EC are proven to be based on the same essential form but derived with different numerical algorithms. We show that the DC framework successfully unifies the theories of GC, TE and EC and their conditional versions, i.e. conditional GC (cGC), conditional TE (cTE) and conditional EC (cEC). This DC framework will provide not only mathematical insights into causality but also clearer understanding of existing causal inference approaches to further explore the concept of causes and effects. Moreover, based on the DC framework, we propose new criteria called embedding entropy (EE) and conditional EE (cEE) to measure the causality and direct causality between variables, respectively. EE and cEE can detect causality based only on measured data, which not only carries the advantages of traditional methods but also simultaneously solves their causal inference problems in terms of nonlinearity and non-separability. Theoretically, we show that the computational results of this approach can ensure causal relations, provided that the measured system is inherently governed by a dynamical system on a general attractive manifold. The results of numerical simulations demonstrate the validity and robustness of EE and cEE, and furthermore, causal inference on various real-world datasets validates their good performance and effectiveness. Hence, EE and cEE are promising as universal metrics for detecting DC in complex networks generally.

We summarize the purposes of this study as follows: (i) giving the definition of the DC and further showing that GC/TE/EC and their conditional versions are different numerical realizations of this concept and (ii) proposing the index EE/cEE to overcome the nonlinearity, non-separability and scale-bias problems in numerical causality inference.

2. Methods

2.1. Dynamical causality framework

To define DC, we begin with the following delayed autonomous dynamical system:

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{f}(\boldsymbol{x}(t), \boldsymbol{x}(t-\theta_1), \boldsymbol{x}(t-\theta_2), \ldots, \boldsymbol{x}(t-\theta_K)) + \boldsymbol{\sigma}_t^{\text{in}}, \qquad (2.1)$$

where the state $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)^T$ contains n variables, $\boldsymbol{f} = (f_1, f_2, \ldots, f_n)^T$ are smooth functions, and $\theta_K > \cdots > \theta_2 > \theta_1 > 0$ are K time delays. Intrinsic noise $\boldsymbol{\sigma}_t^{\text{in}}$ with mean zero is independent of $\boldsymbol{x}$. We assume that the steady states of the system are constrained on a low-dimensional attractive manifold, which generally holds for most real-world dissipative systems. Equation (2.1) is a common dynamical model for time-series data. After integration, time truncation, and numerical discretization (see electronic supplementary material, SI §I), equation (2.1) is transformed into

$$\boldsymbol{x}(t) = \widehat{\boldsymbol{F}}[\boldsymbol{x}(t-1), \boldsymbol{x}(t-2), \ldots, \boldsymbol{x}(t-p)] + \boldsymbol{\epsilon}_t, \qquad (2.2)$$

where $\widehat{\boldsymbol{F}}$ denotes functions in finite-dimensional space to approximate the evolution operator, $\{\boldsymbol{x}(t-1), \boldsymbol{x}(t-2), \ldots, \boldsymbol{x}(t-p)\}$ are p sampled points/states along the trajectory of $[t-\tau, t)$, $\tau = p$ is referred to as the 'memory time', and $\boldsymbol{\epsilon}_t$ is an error term encompassing the intrinsic noise, the truncation error and the discretization error. Independent extrinsic noise with a constant mean over time can be added to the dynamics, but it does not influence the theoretical analysis in the following. For the simplification of notations, we use the step size $\Delta t = 1$ in the main text, while a changeable version with memory time $\tau = p\Delta t$ is discussed in the electronic supplementary material, SI §I. In equation (2.2), only the influence within time τ before t is considered, which is consistent with the fact that the longer τ is, the more causal effects there are. We present a detailed discussion of τ in the electronic supplementary material, SI §I. Based on equation (2.2) and the assumption that $\boldsymbol{\epsilon}_t$ is independent of $\boldsymbol{x}$ with a constant mean over time, discrete DC is defined to investigate the dependence between variables under the function $\widehat{\boldsymbol{F}}$ as follows.

Definition 2.1. The **discrete DC** from x_j to x_i in equation (2.2) is generally defined as $\exists k \in \{1, 2, \ldots, p\}$, such that $\partial \widehat{F}_i / \partial x_j(t-k) \neq 0$ for almost any sampled t. Otherwise, x_j has no discrete DC to x_i.

Here, 'generally' means that there is no generalized synchrony in the system (see [27,28] and electronic supplementary material, SI §I).

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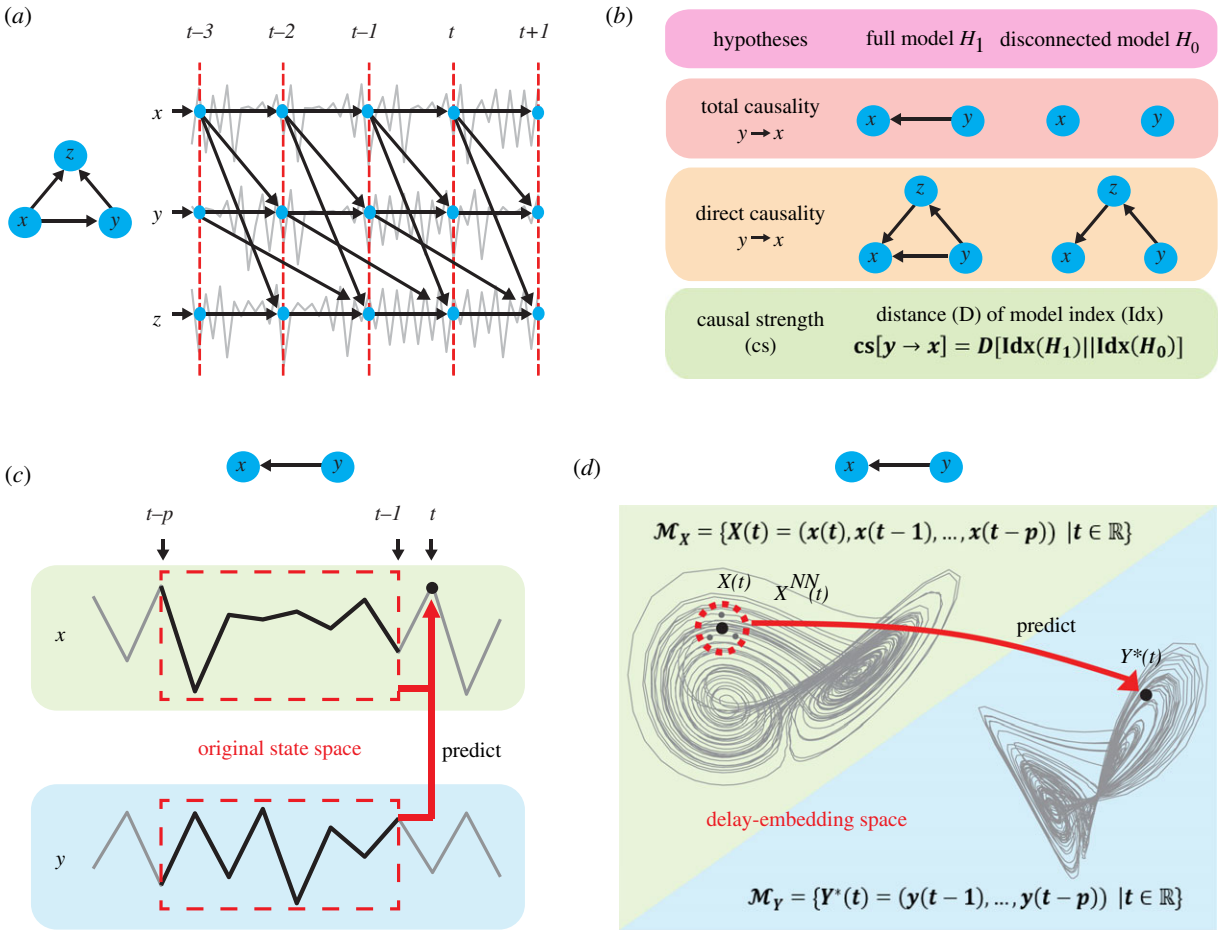


Figure 1. Framework of dynamical causality (DC). (*a*) Illustration of discrete DC, where *x* influences both *y* and *z* with time delay 1, and *y* has the causality to *z* with time delay 2. *x* has not only direct DC to *z* but also indirect DC to *z* through *y*. (*b*) List of basic elements of the DC framework, including the full model, disconnected model, and measurement of causal strength. (*c*) Principle of Granger causality (GC) and transfer entropy (TE) when detecting DC from *y* to *x*. GC and TE use the historical series in the original state space (time-series space) to predict $x(t)$ and measure the predictability. (*d*) Principle of embedding causality (EC) and embedding entropy (EE) when detecting DC from *y* to *x*. EC and EE use Takens' embedding mapping and implicit function theorem in the delay embedding space and measure the predictability from delay embedding manifold $\mathcal{M}_X$ to $\mathcal{M}_Y$. In the corresponding numerical algorithms, the neighbours $X^{NN}(t) \in \mathcal{M}_X$ around $X(t)$ are used to predict $Y^*(t) \in \mathcal{M}_Y$. Note that there is no $y(t)$ element in $Y^*(t)$, but there is an $x(t)$ element in $X(t)$.

In fact, discrete DC can also be called **discrete τ-time global DC** due to the limited time trajectory during $[t-\tau, t)$ for any state $\boldsymbol{x}$. We use the term **DC** for convenience in the following. **Direct DC** corresponds to the explicit dependence of $\widehat{F}_i$ on x_j, whereas **indirect DC** entails the implicit dependence of $\widehat{F}_i$ on x_j. We also provide a theoretical definition for DC in the continuous case in the electronic supplementary material, SI §I. Figure 1*a* illustrates the concept of DC.

To quantify the strength of causality, the following two models are usually considered [29]:

— The **full model** (H_1 model/hypothesis) assumes that the causality under consideration truly exists (i.e. x_j has DC to x_i, $x_j \to x_i$). To focus on the DC from y to x, a simplified full model is defined as follows:

$$\begin{cases} x(t) = f(x(t-1), \ldots, x(t-p), y(t-1), \ldots, y(t-p)) + \epsilon_{x,t}, \\ y(t) = g(y(t-1), \ldots, y(t-p)) + \epsilon_{y,t}. \end{cases} \tag{2.3}$$

— The **disconnected model** (H_0 model/hypothesis) assumes that the causality under consideration vanishes (i.e. x_j does not have DC to x_i, while other relations in the system remain unchanged from the full model, $x_j \nrightarrow x_i$). To focus on the DC from y to x, a simplified disconnected model is defined as follows:

$$\begin{cases} x(t) = f(x(t-1), \ldots, x(t-p)) + \epsilon_{x,t}, \\ y(t) = g(y(t-1), \ldots, y(t-p)) + \epsilon_{y,t}. \end{cases} \tag{2.4}$$

The equation of y could either depend on x or not, which does not affect the causation of x. Clearly, according to the definition of DC, $x(t)$ depends on some $y(t-k)$ in the H_1 model but does not rely on any $y(t-k)$ in the H_0 model, where $k = 1, 2, \ldots, p$. Let $\mathrm{Idx}(H_i)$ be some well-defined index under H_i, e.g. entropy in TE or PCC in EC. Then, the strength of DC can be defined as a distance between the two models' indices as follows.

Definition 2.2. The strength of DC from x_j to x_i is defined as

$$\mathrm{cs}[x_j \to x_i] = D[\mathrm{Idx}(H_1) \| \mathrm{Idx}(H_0)], \tag{2.5}$$

where cs denotes the causal strength, $D(\cdot \,||\, \cdot)$ is some distance function, and H_0 and H_1 are the corresponding models with and without DC from x_j to x_i, respectively.

Figure 1*b* summarizes the concepts of hypotheses and causal strength of the DC framework.

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For the numerical computation, DC is assumed to be evaluated on the steady attractive manifold of a non-synchronized system, where statistical properties can be calculated from the data.

2.2. Traditional causality algorithms under the dynamical causality framework

GC, TE and EC are the most prevalently employed algorithms for detecting causality between variables. In the following, we will use equations (2.3) and (2.4) as examples to illustrate the consistency of GC, TE and EC with the DC framework. Unless otherwise specified, DC is considered from y to x. The time label t will be denoted as either $x(t)$ or x_t.

GC uses the vector autoregression technique to detect whether the cause variable appears in the evolution equation of the effect variable. Variation is used to compare the full and disconnected models. Specifically, as shown in table 1, GC uses the linearized equations (2.3) and (2.4) as the H_1 model with y and the H_0 model without explicit y to predict x, respectively. Its model index is $-\ln\mathrm{Var}(\epsilon_i)$, where ϵ_i is the variation of the residue of variable x in the original state space (x_t, y_t); thus, the causal strength of GC is $\mathrm{GCI}_{y\to x} = |-\ln\mathrm{Var}(\epsilon_1) + \ln\mathrm{Var}(\epsilon_2)|$. In other words, if the residue ϵ for predicting x with y becomes smaller than that without explicit y, then $y \to x$ in the GC sense.

TE generalizes GC to nonlinear cases by using entropy to measure the information transferred between variables, still in the original state space (x_t, y_t). In other words, as shown in table 1, TE considers the nonlinear forms of equations (2.3) and (2.4) as the H_1 model with y and the H_0 model without explicit y, respectively. Its model index is the entropy $H(x_t \mid x_{t-1}, \ldots, x_{t-p}, y_{t-1}, \ldots, y_{t-p})$, which avoids the direct construction of evolving operator, and the causal strength of TE is $\mathrm{TE}_{y\to x} = |H(x_t \mid x_{t-1}, \ldots, x_{t-p}, y_{t-1}, \ldots, y_{t-p}) - H(x_t \mid x_{t-1}, \ldots, x_{t-p})|$. In other words, if the uncertainty of x with y becomes smaller than that without explicit y, then $y \to x$ in the TE sense.

By contrast, EC considers causality in the delay embedding space and uses the embedding mapping to realize DC. It solves the so-called non-separability problem encountered by GC and TE [15,26]. We provide a detailed discussion of the non-separability in the electronic supplementary material, SI §II(D). EC considers the dynamics of $X_t = (x(t), x(t-1), \ldots, x(t-p))^T \in \mathcal{M}_X$ and $Y_t^* = (y(t-1), \ldots, y(t-p))^T \in \mathcal{M}_Y$ in the delay embedding space, where $\mathcal{M}_X$ and $\mathcal{M}_Y$ are the corresponding attractive manifolds of X_t and Y_t^*, respectively. Takens' embedding theorem ensures embedding mapping [30–32] from $\mathcal{M}_X$ to $\mathcal{M}_Y$ in H_1, which is just the implicit function of Y_t^* contained in the dynamics of X_t. EC uses the local linear cross-mapping technique to obtain prediction/reconstruction from $\mathcal{M}_X$ to Y_t^*, denoted as $\widehat{Y}_t$. Here, the $p+2$ neighbours of X_t are cross-mapped to $Y_{t_{x_i}}^*$ with $i = 1, \ldots, p+2$, whose linear combination with weight w_i is further used to predict $\widehat{Y}_t = \sum_{i=1}^{p+2} w_i Y_{t_{x_i}}^*$. The model index is the Pearson correlation coefficient $\mathrm{PCC}(\widehat{Y}_t, Y_t^*)$ (the mean value of the component-wise PCCs), and the causal strength of EC is $\mathrm{EC}_{y\to x} = |\mathrm{PCC}(\widehat{Y}_t, Y_t^* | H_1)|$ with $\mathrm{PCC}(\widehat{Y}_t, Y_t^* | H_0) = 0$ for H_0. Instead of using data to construct the H_0 model, EC sets the index under H_0 to zero by default. Table 1 gives a summary of EC. Intuitively, if the neighbours of X_t on $\mathcal{M}_X$ can be local-linearly cross-mapped to reconstruct Y_t^* on $\mathcal{M}_Y$, then $y \to x$ in the EC sense.

For a short summary, figure 1*c* illustrates GC and TE, which detect DC in the original state space, and figure 1*d* illustrates EC, which identifies DC in the delay embedding space. Table 1 summarizes their properties, specifically listing their H_0 models, H_1 models, and model indices based on the DC framework. GC, TE, and EC are simply different realizations of the DC framework from a dynamical viewpoint. These results can also be generalized to their conditional versions, i.e. cGC, cTE and cEC. Detailed descriptions can be found in the electronic supplementary material, SI §II and §III.

2.3. Embedding entropy and conditional embedding entropy

To combine the advantages of TE/cTE and EC/cEC, we propose two indices under our DC framework, called EE and cEE, which can deal with both nonlinear time series and non-separable variables.

For the case of two variables x and y, from equation (2.2), we have the H_1 model as in equation (2.3). Using EC notations, we transform the original state space to the delay embedding space as $X(t) = (x(t), x(t-1), \ldots, x(t-p))^T \in \mathbb{R}^{p+1}$ and $Y^*(t) = (y(t-1), \ldots, y(t-p))^T \in \mathbb{R}^p$. We know that $(X(t)^T, Y^*(t)^T) \in \mathbb{R}^{2p+1}$ is included in a low-dimensional $\mathbb{R}^L$ manifold based on the assumption of steady dynamics, where L is the inner dimension of the system. Takens' embedding theorem [30–32] ensures the implicit function

$$Y^*(t) = \boldsymbol{h}(X(t), \boldsymbol{\epsilon}_{x,t}) \tag{2.6}$$

from equation (2.3), where $\boldsymbol{h}$ is the embedding mapping from manifold $\mathcal{M}_X = \{X(t) | t \in \mathbb{R}\}$ to $\mathcal{M}_Y = \{Y^*(t) | t \in \mathbb{R}\}$. Let $X^{NN}(t) = (X^{[1]}(t), X^{[2]}(t), \ldots, X^{[p+2]}(t))$, where $X^{[k]}(t)$ is the kth NN (nearest neighbour) of $X(t)$ on $\mathcal{M}_X$. EC constructs $\boldsymbol{h}$ by the linear combination of local neighbours in $X^{NN}(t)$ (i.e. $\sum_{i=1}^{p+2} w_i Y_{t_{x_i}}^*$), while we avoid the direct construction of $\boldsymbol{h}$, but measure DC by the nonlinear mutual information $\mathrm{MI}(Y^*, X^{NN})$.

Definition 2.3. The causal strength of embedding entropy (EE) from variable y to x is defined as

$$\mathrm{EE}[y \to x] = \mathrm{MI}(Y^*, X^{NN}), \tag{2.7}$$

where $\mathrm{MI}(\cdot, \cdot)$ is the mutual information.

EE can be considered as the endogenous information of $\mathcal{M}_Y$ contained in $\mathcal{M}_X$ under H_1, which is zero as default under H_0 because Y^* is independent of X^{NN}. In summary, EE combines the entropy in TE to overcome nonlinearity and the delay embedding mapping in EC to address non-separability. Taking mutual information as the model index, EE also satisfies our DC framework. Note that $Y^*(t)$ has no element $y(t)$, which maintains the directionality of the causal index.

The same derivation is applicable to any multi-variable case, e.g. for x, y and z. Thus, we have cEE to detect conditional/direct DC as follows.

Definition 2.4. Conditional embedding entropy from y to x conditioned on z has the expression

$$\mathrm{cEE}[y \to x | z] = \mathrm{CMI}(Y^*, X^{NN} | Z^{NN}), \tag{2.8}$$

where $\mathrm{CMI}(\cdot, \cdot | \cdot)$ is the conditional mutual information.

In equation (2.8), $Z^{NN}(t)$ contains the $p+2$ NNs of $Z(t) = (z(t), z(t-1), z(t-2), \ldots, z(t-p))^T \in \mathbb{R}^{p+1}$. $\mathrm{CMI}(\cdot, \cdot | \cdot)$ is used as the model index, and it is set as zero under H_0.

EE and cEE are also suitable for detecting causality between groups containing one or several variables in either cause or effect. The major features of EE/cEE are (1) the use of entropy to measure nonlinear causality without explicit mapping construction by adopting information between variables as the model index, (2) the use of embedding mapping to represent the implicit function from Y^* to X, which solves the non-separability problem, and (3) the use of X^{NN} instead of X in the delay embedding space to avoid the false negative problem (scale bias) induced by strong correlations [33,34]. Figure 1*d* also illustrates the concept of EE, and table 1 summarized its properties. More details and remarks for EE/cEE can be found in the electronic supplementary material, SI §IV.

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Table 1. Summary of dynamical properties of Granger causality (GC), transfer entropy (TE), embedding causality (EC) and embedding entropy (EE).

	GC	TE	EC	EE
modelling space	original state space (x_t, y_t)	original state space (x_t, y_t)	delay embedding space (X_t, Y_t^*)	delay embedding space (X_t, Y_t^*)
mapping	linear	nonlinear	local linear	nonlinear
solvability of non-separability problem	no	no	yes	yes
numerical algorithm	vector autoregression	transfer entropy	convergent cross mapping	embedding entropy
H_1 model for $y \to x$ (the full model)	$x_t = \sum_{i=1}^{p} a_i x_{t-i} + \sum_{i=1}^{p} b_i y_{t-i} + \epsilon_{1,t}$	$x_t = f(x_{t-1}, \ldots, x_{t-p}, y_{t-1}, \ldots, y_{t-p}) + \epsilon_{1,t}$	$\widehat{Y}_t = \sum_{i=1}^{p+2} w_i Y^*_{t_{x_i}}$ can positively cross-reconstruct Y_t^* from X_t neighbours (X_t^{NN}), i.e. $\mathrm{PCC}(\widehat{Y}_t, Y_t^* \vert H_1) > 0$	$Y_t^* = \boldsymbol{h}(X_t^{NN}, \epsilon_{x,t})$ and X_t neighbours X_t^{NN} contain non-zero information of Y_t^*, i.e. $\mathrm{MI}(Y_t^*, X_t^{NN} \vert H_1) > 0$
H_0 model for $y \nrightarrow x$ (the disconnected model)	$x_t = \sum_{i=1}^{p} a_i x_{t-i} + \epsilon_{2,t}$	$x_t = f(x_{t-1}, \ldots, x_{t-p}) + \epsilon_{2,t}$	$\widehat{Y}_t$ cannot cross-reconstruct Y_t^* from X_t neighbours, i.e. $\mathrm{PCC}(\widehat{Y}_t, Y_t^* \vert H_0) = 0$	X_t neighbours X_t^{NN} contain no information of Y_t^*, i.e. $\mathrm{MI}(Y_t^*, X_t^{NN} \vert H_0) = 0$
model index	$-\ln(\mathrm{Var}(\epsilon_i))$	$H(x_t \vert x_{t-1}, \ldots, x_{t-p}, y_{t-1}, \ldots, y_{t-p})$	$\mathrm{PCC}(\widehat{Y}_t, Y_t^*)$	$\mathrm{MI}(Y_t^*, X_t^{NN})$
causal strength (distance between indices)	$\mathrm{GCI}_{y\to x} = \lvert -\ln\mathrm{Var}(\epsilon_1) + \ln\mathrm{Var}(\epsilon_2) \rvert$	$\mathrm{TE}_{y\to x} = \lvert H(x_t \vert x_{t-1}, \ldots, x_{t-p}, y_{t-1}, \ldots, y_{t-p}) - H(x_t \vert x_{t-1}, \ldots, x_{t-p}) \rvert$	$\mathrm{EC}_{y\to x} = \lvert \mathrm{PCC}(\widehat{Y}_t, Y_t^* \vert H_1) \rvert$	$\mathrm{EE}_{y\to x} = \lvert \mathrm{MI}(Y_t^*, X_t^{NN} \vert H_1) \rvert$

In the table, $X_t = (x_t, x_{t-1}, \ldots, x_{t-p})^T$ with x_t element, while $Y_t^* = (y_{t-1}, y_{t-2}, \ldots, y_{t-p})^T$ without y_t element. X_t^{NN} represents the nearest neighbours of X_t in the delay embedding space. The subscripts of $Y^*_{t_{x_i}}$ with $i = 1, \ldots, p+2$ are the time labels corresponding to X_t's $p+2$ nearest neighbours (X_t^{NN}) in the delay embedding space. $\mathrm{Var}(\cdot)$ represents the statistical variation, $H(\cdot)$ is the entropy, $\mathrm{PCC}(\cdot, \cdot)$ is the Pearson correlation coefficient, and $\mathrm{MI}(\cdot, \cdot)$ is the mutual information. cGC, cTE, cEC and cEE have similar properties but with an additional conditioned variable.

2.4. Numerical algorithms

For the numerical computation, GC/cGC were realized by the vector auto-regression. The estimations of MI and CMI in TE/cTE/EE/cEE were calculated by the kNN approach [35], which originates from the Kozachenko–Leonenko strategy [36] and is suitable for high-dimensional variables. The kNN estimator has been a standard method for TE/cTE in toolboxes such as TRENTOOL [37]. EC/cEC were realized by the convergent cross mapping (CCM) algorithm [15,28] and the partial cross mapping (PCM) algorithm [26], respectively. EE/cEE were realized by the numerical MI and CMI according to equations (2.7) and (2.8) in the delay-embedding space, respectively. More details are provided in the electronic supplementary material, SI §V.

There are two goals (detecting total causality and direct causality) and three problems (nonlinearity, non-separability and scale bias) in the numerical calculation, whose detailed description is shown in the electronic supplementary material, SI §V(C). EE/cEE are proposed to reach the two goals and overcome the three problems, which hinder the wide application of traditional methods.

3. Results

3.1. Effectiveness and robustness of EE/cEE on simulated datasets

In the following, we will describe the numerical properties of EE/cEE on four simulated datasets: a two-dimensional system with unidirectional/bidirectional couplings, a three-dimensional discrete logistic map, a continuous coupled Lorenz system and a multi-variable coupled Henon map. The comparisons with GC/cGC, TE/cTE and EC/cEC are also presented. As there is no well-defined criterion of convergence in measuring the causal strength, we used the *p*-value and the ROC/AUC to evaluate the performance of different indices in this study, following the convention of statistics. Detailed dynamics for the simulated systems can be found in the electronic supplementary material, SI §VI. These simulated examples show that EE/cEE can improve the accuracy, hold the robustness and decrease the scale bias over traditional indices.

3.1.1. Unidirectional/bidirectional couplings

We first compared GC, TE, EC and EE by a two-dimensional coupling system (see electronic supplementary material, SI §VI(A)). x and y are two variables. β_{xy} adjusts the causality from x to y, and β_{yx} is the coupling coefficient from y to x. The unidirectional case is set as $\beta_{yx} = 0$, and the values for the four indices are shown in figure 2*a*,*b* with β_{xy} ranging from 0 to 0.45. The bidirectional case with $\beta_{yx} = 0.1$ is shown in figure 2*c*,*d*. This simulated dataset shows that detecting causality from the delay-embedding space (EC/EE) is more accurate than that in the original space (GC/TE).

First, GC is not suitable for the nonlinear and non-separable system. In figure 2*a*,*c*, GC[$x \to y$] does not monotonically increase with β_{xy}, and it is far from 0 even when $\beta_{xy} = 0$. In figure 2*b*,*d*, though β_{yx} keeps constant, GC[$y \to x$] varies largely.

Second, TE suffers from the false negative problem, and cannot tell the existing causality. When comparing the causality from y to x under the H_0 model in figure 2*b* (no edge from y to x) with the H_1 model in figure 2*d* (an edge from y to x), we measured the difference of distributions at $\beta_{xy} =$ 0.1 with the Wilcoxon ranksum test. The *p*-values ($p_{GC} = 2.20 \times 10^{-10}$, $p_{TE} = 3.25 \times 10^{-1}$, $p_{EC} = 2.56 \times 10^{-34}$, and $p_{EE} = 3.29 \times 10^{-19}$) support the conclusion.

Third, EC/EE are less susceptible than GC/TE by the other couplings. By comparing figure 2*a*,*c*, we used the Wilcoxon ranksum test to check if the indices $x \to y$ have a significant change influenced by the opposite coupling $y \to x$. Under the null hypothesis that an index has the same value at $\beta_{yx} = 0$ and $\beta_{yx} = 0.1$, electronic supplementary material, table S2, shows the *p*-values. We can find that GC/TE suffer from the influence of the opposite coupling, because they fit an H_0 model with the data from H_1 and are not suitable for the non-separable system.

In addition, we tested the performance of indices under different threshold values (electronic supplementary material, figure S2*a*–*c*), which shows that when the threshold value is in [0.001, 0.058], EE has the least false rate among all indices (electronic supplementary material, SI §VI(A)).

3.1.2. Discrete logistic map

Next, we considered a three-dimensional discrete logistic map to demonstrate the numerical performances of the eight DC criteria from x to z (see electronic supplementary material, SI §VI(B)). When the associations between x and z and those between y and z have moderate strength ($\beta_{xy} = 0.2$, $\beta_{yz} = 0.2$), we change the direct causality from x to z by β_{xz}, from 0 to 0.45. The values for the eight indices can be seen in figure 3*a*,*b*. A comparison of H_0 and weak H_1 ($\beta_{xz} = 0.1$) is shown in figure 3*c*,*d*. When we strengthen the relationship between x and z and between y and z with $\beta_{xy} = \beta_{yz} = 0.45$, the performance of weak H_1 ($\beta_{xz} = 0.1$) is plotted in figure 3*e*. The detailed mean values and standard deviations of the indices are listed in electronic supplementary material, table S1. This simulated dataset shows that EE/cEE can overcome the scale bias and be more accurate in detecting total/directed causality than other indices.

First, TE is prone to be false negative even if there exists total causality. When $\beta_{xz} = 0$ in figure 3*a*, GC/TE/EC/EE should always be larger than 0 because indirect causality from x to z through y always exists. However, TE is prone to suffer from the false negative due to its scale bias (figure 3*c* and the first column in electronic supplementary material, table S1). EE can measure the total causality correctly.

Second, cEE has less scale bias in detecting direct causality. When $\beta_{xz} = 0$ in figure 3*b*, there is no direct causality from x to z, but cGC/cEC are prone to be false positive (figure 3*c* and the first column in electronic supplementary material, table S1). When $\beta_{xz} = 0.1$ in figure 3*b*, there is always direct causality from x to z, but cTE is prone to be false negative (figure 3*d* and the second column in electronic supplementary material, table S1).

Third, cEE is less susceptible than cGC/cEC by the indirect couplings. Theoretically, strengthening the indirect couplings β_{xy} and β_{yz} increases the total causality between x and z, but should not change the direct causality between them. Under the null hypothesis that an index in figure 3*d* has the same medial value as in figure 3*e*, we use the Wilcoxon ranksum test and find that the indices GC/TE/EC/EE correctly have significant differences in detecting total causality ($p_{GC} = 1.21 \times 10^{-32}$, $p_{TE} = 5.80 \times 10^{-31}$, $p_{EC} = 2.72 \times 10^{-34}$,

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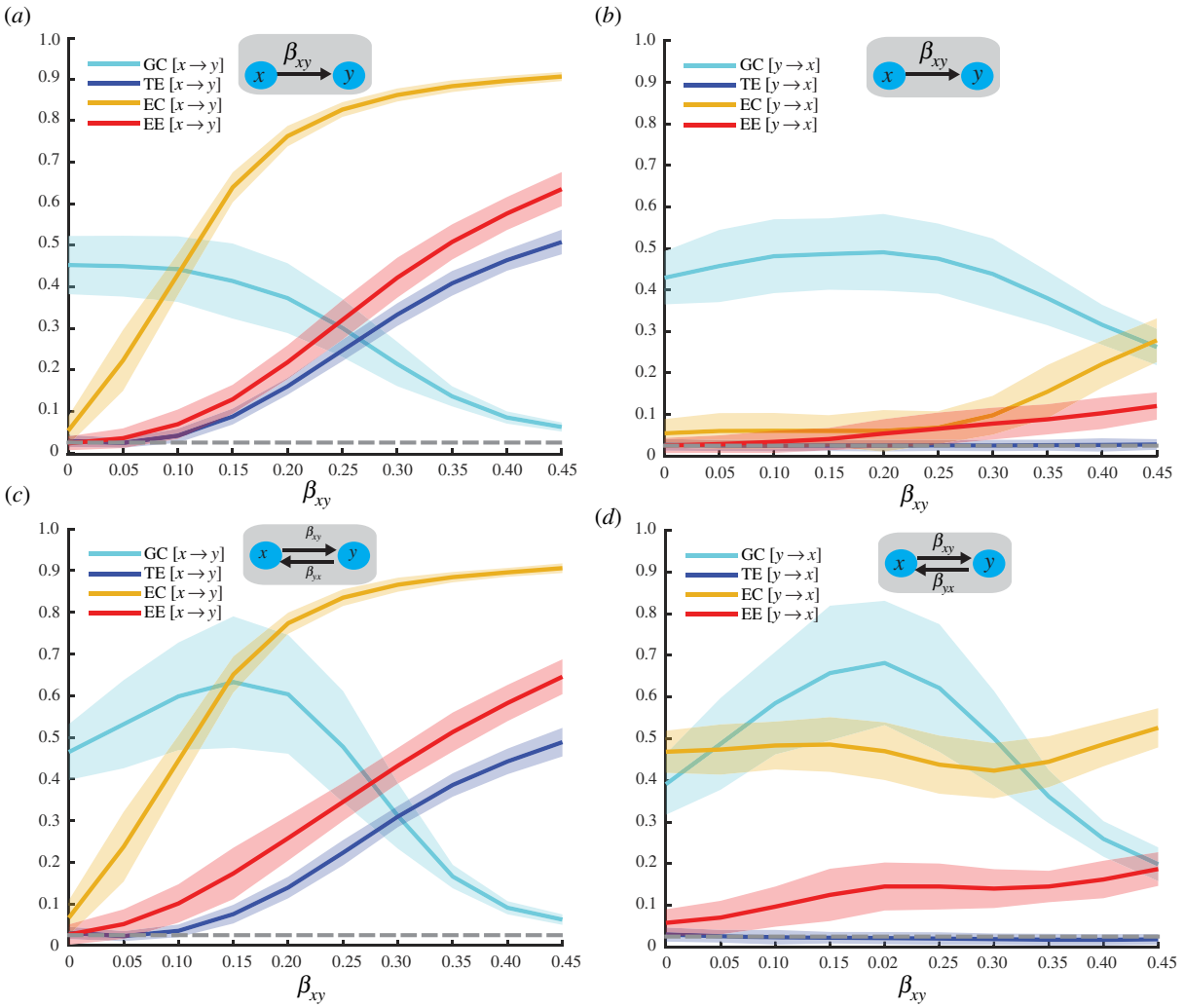


Figure 2. Comparison of four criteria for dynamical causality on the unidirectional and bidirectional couplings. Parameter β_{xy} varies from 0 to 0.45, which corresponds to the strength of coupling from x to y. β_{yx} is the strength of coupling from y to x, which is constant in each panel. (*a*,*b*) The unidirectional case with $\beta_{yx} = 0$; (*c*,*d*) the bidirectional case with $\beta_{yx} = 0.1$. (*a*,*c*) The mean values (lines) and standard derivations (shaded) of GC, TE, EC and EE from x to y. (*b*,*d*) The corresponding indices from y to x. 100 time series were generated for each parameter to estimate the standard derivation. The grey lines in each panel are the reference of 0.025.

$p_{EE} = 2.56 \times 10^{-34}$). However, only cTE and cEE correctly show the insignificant changes in detecting the direct causality with $p_{cTE} = 2.62 \times 10^{-1}$ and $p_{cEE} = 2.07 \times 10^{-1}$, respectively. cGC and cEC show significant increase with $p_{cGC} = 2.49 \times 10^{-11}$ and $p_{cEC} = 1.13 \times 10^{-28}$, respectively.

In addition, we changed the threshold value from 0 to 1 (electronic supplementary material, figure S2*d–f*), and found that cEE has the least average false rate among the indices when the threshold value is in [0.014, 0.035] (electronic supplementary material, SI §VI(B)). From the computation in electronic supplementary material, figure S2*g*,*h*, we also conclude that only cEE can distinguish the direct DC $x \to z$ from the opposite edge $z \to x$ correctly (electronic supplementary material, SI §VI(B)).

The good performances of EE/cEE in figures 2 and 3 can be attributed to three key points: using entropy to solve the nonlinearity, using delay embedding mapping to solve the non-separability and using nearest neighbours on the attractive manifold to solve the scale bias problem.

3.1.3. Continuous coupled Lorenz system

Next, we considered a continuous coupled Lorenz system (see electronic supplementary material, SI §VI(C)) to test the robustness of indices from x to z under different intrinsic noise perturbations, extrinsic noise perturbations, orders p and discrete step sizes Δt (sample density).

The causality to be detected is between groups (x_1, x_2, x_3) and (z_1, z_2, z_3), while (y_1, y_2, y_3) is the medial/conditional group (figure 4*a*). The robustness of EE/cEE under intrinsic and extrinsic perturbations is shown in figure 4*c–f*. In electronic supplementary material, figures S3 and S4, we also compare EE/cEE with six other indices, which shows that EE/cEE is as stable as other criteria under intrinsic noise and decreases slower than the other indices under increasing extrinsic noise. We also analyse different orders p with fixed step Δt and different steps Δt with fixed memory time τ (i.e. $p = \tau/\Delta t$ with different sample densities in a unit time interval), and the results are shown in electronic supplementary material, figures S5 and S6, respectively. These figures, along with the ANOVA results in electronic supplementary material, SI §VIII(D), indicate that EE/cEE are stable and robust for different parameters.

3.1.4. Multi-variable coupled Henon map

We considered a multi-variable coupled Henon map system to test the indices' performances in constructing a causality

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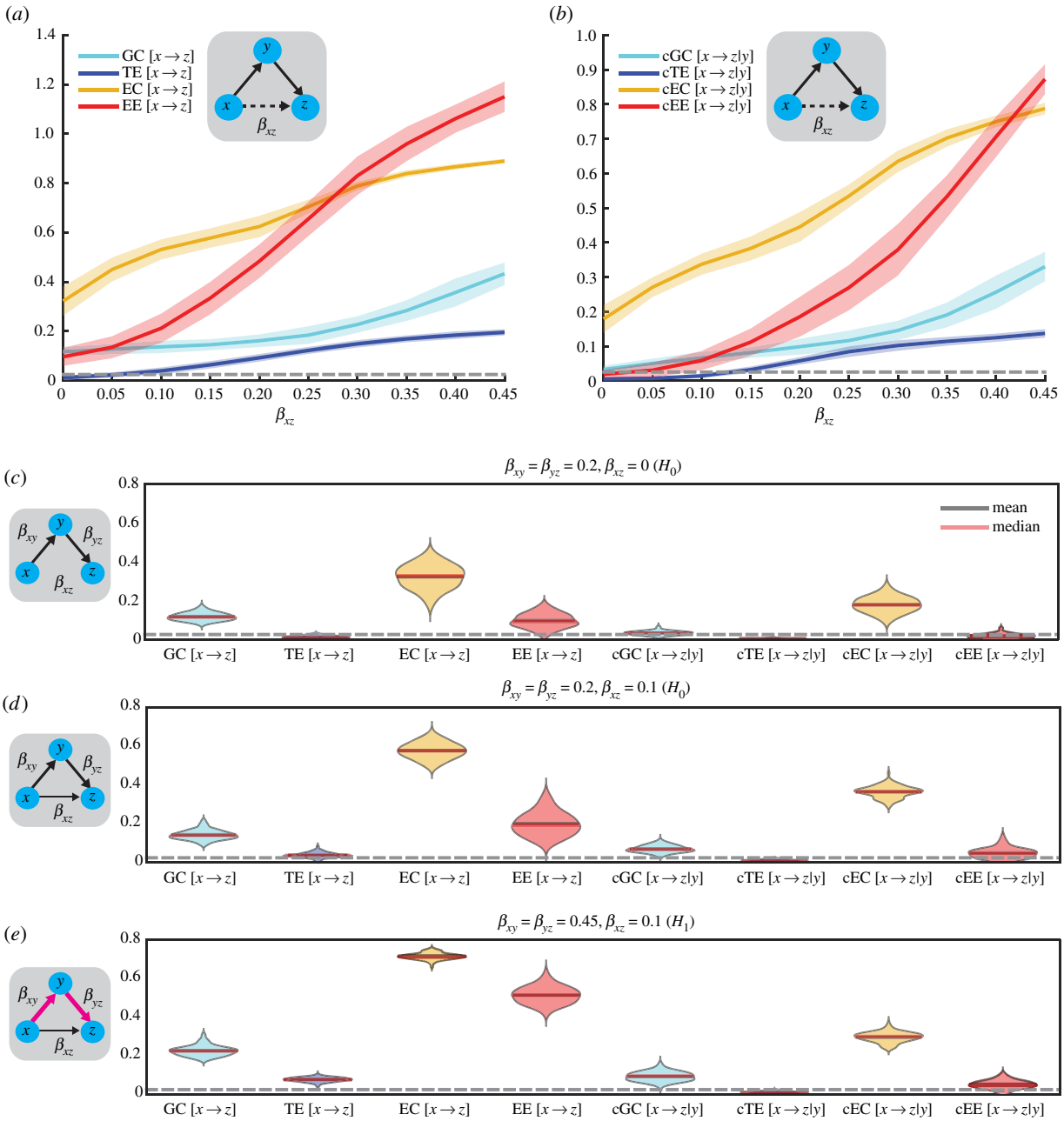


Figure 3. Comparison of eight criteria for dynamical causality on the discrete logistic map. (*a*,*b*) Parameter β_{xz} varying from 0 to 0.45, which corresponds to the strength of direct causality from *x* to *z*. 100 time series were generated for each parameter. (*a*) The mean values (lines) and standard derivations (shaded) of GC, TE, EC and EE from *x* to *z*, which detect the total causality from *x* to *z*, including direct and indirect causality. (*b*) The mean values (lines) and standard derivations (shaded) of cGC, cTE, cEC and cEE from *x* to *z* conditioned on *y*, which only detect the direct causality from *x* to *z*. (*c*–*e*) Violin plots showing the numerical properties of different indices under a moderately correlated network (*c* and *d* with $\beta_{xy} = \beta_{yz} = 0.2$) and strongly correlated network (*e* with $\beta_{xy} = \beta_{yz} = 0.45$). For the direct causality from *x* to *z*, (*c*) shows the H_0 model ($\beta_{xz} = 0$), while (*d*,*e*) show the weak H_1 model ($\beta_{xz} = 0.1$). The grey lines in each panel are reference lines for the value of 0.025.

network (see electronic supplementary material, SI §VI(D)). For the 10-node network in figure 4*b*, we compute the values of 8 indices for each directed edge between every pair of nodes. Figure 4*e*,*f* shows the receiver operating characteristic (ROC) curves and the area under the curve (AUC) values for the 8 indices. EE and cEE have better AUCs than the other indices when detecting total DC and direct DC, respectively. In electronic supplementary material, figure S7, we also plot the networks determined by the 8 indices when using the optimal threshold value determined by the ROC curve (see electronic supplementary material, SI §VI(D)). These results show that EE/cEE are more efficient in inferring multi-variable causality networks than the other indices.

3.2. Performance of EE/cEE on real datasets

EE/cEE are also used to detect the dynamical causality contained in real observed time-series data. We briefly describe the results of three real datasets in the following, whose detailed information can be found in the electronic supplementary material, SI §VII. These real applications indicate that EE/cEE are promising indices to detect dynamical causality in the real world.

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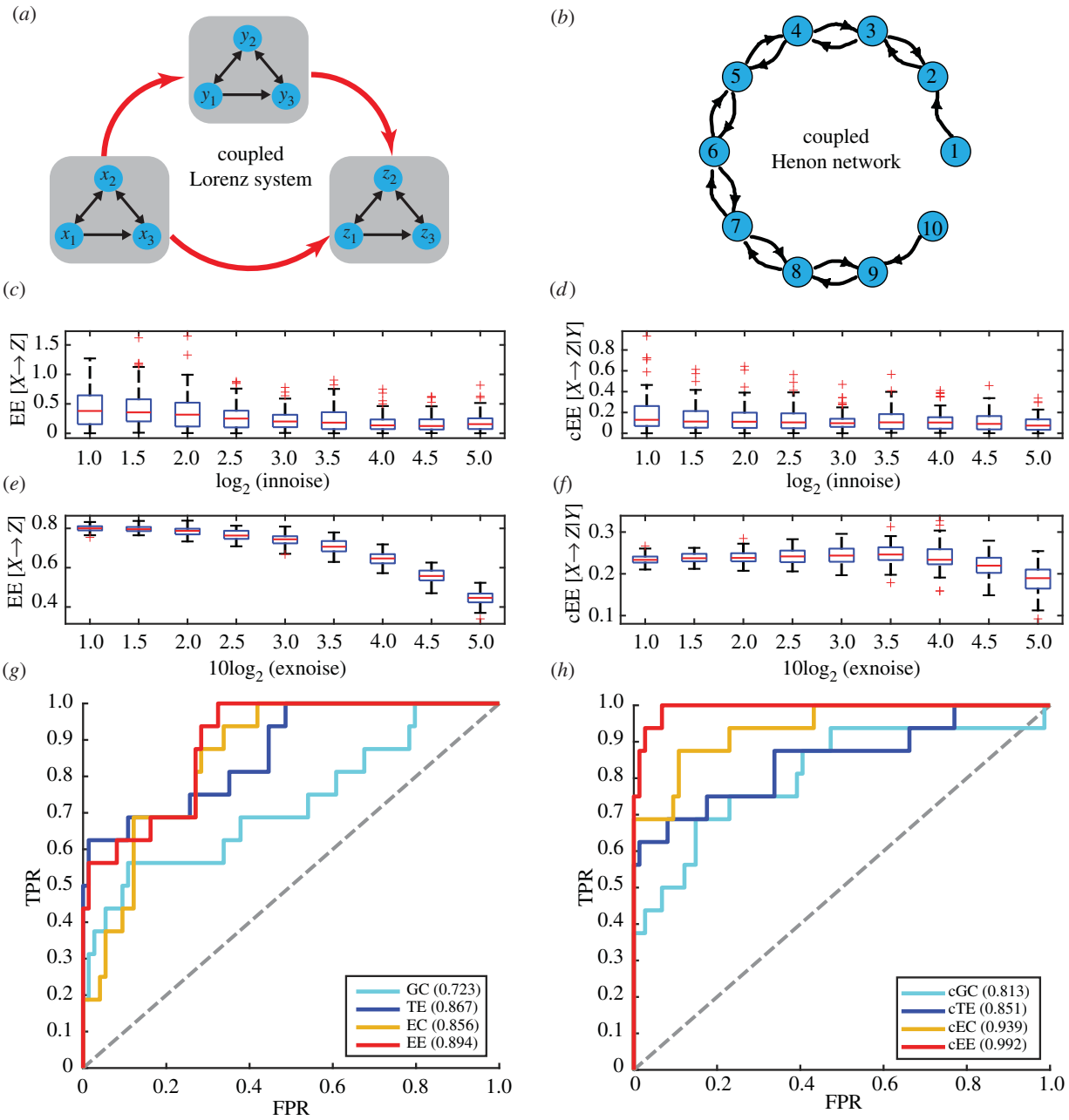


Figure 4. Robustness of EE/cEE in the coupled Lorenz system and its causality inference efficiency in the coupled Henon network. (*a*) Causality between variables of the coupled Lorenz system. We measure the dynamical causality (DC) from groups $X = (x_1, x_2, x_3)$ to $Z = (z_1, z_2, z_3)$, while $Y = (y_1, y_2, y_3)$ are the medial variables. (*b*) A 10-node coupled Henon network. DCs are shown by arrows. (*c*,*d*) Robustness of EE/cEE values when increasing the amplitude of intrinsic noise in the coupled Lorenz system. (*e*,*f*) Robustness of EE/cEE values when increasing the amplitude of extrinsic noise in the coupled Lorenz system. (*g*,*h*) Receiver operating characteristic (ROC) curves for 8 indices when inferring the 10-node coupled Henon map causality network and their values of area under the curve (AUC). EE and cEE achieved the best performances over the other indices when detecting total and direct causality, respectively.

3.2.1. Japanese COVID-19 data

We collected the daily number of new infected COVID-19 cases in Japan over the 304 days from 14 February 2020 to 13 December 2020. There are 47 prefectures in Japan, and we use EE to explore the total DC of the COVID-19 infection between each pair of these areas. Electronic supplementary material, figure S8*b*, shows the heatmap of EE for this dataset. Taking Tokyo as an example, Kanagawa, Osaka, Aichi, Saitama and Chiba are the five most affected prefectures in the order of EE. This can be explained by Kanagawa, Saitama and Chiba being neighbours of Tokyo and the heavy traffic between Tokyo and Osaka/Aichi (figure 5*a*). In electronic supplementary material, figure S8*a*, we also show a curve of the infected population for these six prefectures. Thus, besides showing the practicality of EE, we suggest that decreasing the movement of people is an important countermeasure to limit the spread of COVID-19.

3.2.2. Hong Kong air pollution and cardiovascular diseases

Air pollution is believed to be a major cause of cardiovascular diseases (CVDs). We use the data of concentrations of air pollutants and CVD hospital admissions in Hong Kong to detect their causal relations [38]. Figure 5*b* shows the causal network inferred by cEE, which shows that NO_2, SO_2 and respirable suspended particulates (RSP) are direct

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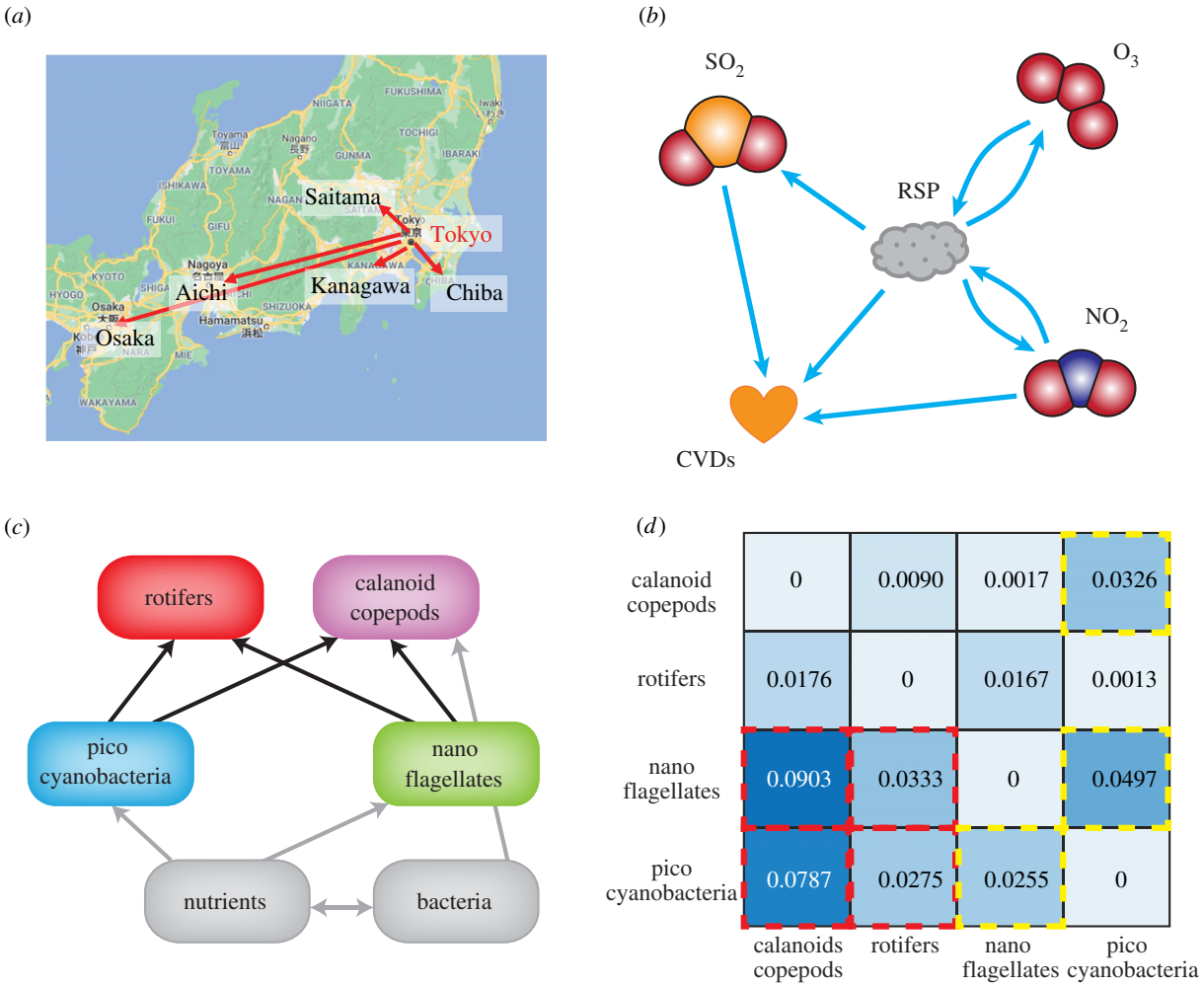


Figure 5. Applications of EE/cEE on real datasets. (*a*) Five prefectures most influenced by Tokyo, calculated using the number of COVID-19 cases in Japan and the EE index: Kanagawa, Osaka, Aichi, Saitama and Chiba. (*b*) The DC network between air pollutes and cardiovascular diseases (CVDs), constructed using the concentrations of air pollutants recorded in Hong Kong and the cEE index. SO_2, respirable suspended particulates (RSP), and NO_2 are considered as direct inducers of CVDs, while O_3 is not. (*c*) The true food chain among rotifers, calanoid copepods, picocyanobacteria and nanoflagellates isolated from the Baltic Sea. Nutrients and bacteria are hidden variables that were not recorded. (*d*) Heatmap of cEE between four species in the food chain dataset. Red squares are true positive causalities from (*c*), while yellow squares are false positive causalities caused by hidden variables.

DCs of CVDs, while O_3 is not (see numerical details in the electronic supplementary material, SI §VII(B)). This is consistent with prior research [39,40]. Electronic supplementary material, figure S9, also shows a comparison with the EE results, which shows that cEE can eliminate many indirect causal conflicts and distinguish the important direct causal factors.

3.2.3. Food chain network

We use the data of a food chain network recorded from the Baltic Sea [41,42] over 1032 days. Four species are considered: calanoid copepods, rotifers, nanoflagellates and picocyanobacteria. A part of the food chain is shown in figure 5*c* (the complete version is shown in electronic supplementary material, figure S10*a*, reproduced from [41]). The dynamical causalities calculated by cEE are shown by the heatmap in figure 5*d*. By choosing a suitable threshold value, i.e. 0.025, from the heatmap, we can see that the producers (nanoflagellates and picocyanobacteria) have strong causal effects on the consumers (calanoid copepods and rotifers), as shown by the red squares in figure 5*d*. There are also causalities which are false positives (yellow squares) but cannot be distinguished from the available data, as there are unobserved hidden common causes (nutrients and bacteria). Electronic supplementary material, figure S10*b*, also shows the result of EE, which suffers more from false positives than cEE.

4. Discussion

In this paper, we proposed the concept of DC based on dynamical systems theory. We proved that the prevalently employed GC, TE and EC are consistent with the DC framework when only considering causes and effects within a limited time range of observed data. The GC, TE and EC are simply different numerical realizations of DC when measuring causal strength from discrete time series samples. We also addressed the two goals (the total causality and direct causality) and three problems (the nonlinearity, non-separability and scale bias) in the causal inference. Especially, the non-separability problem emphasizes that non-separable variables cannot be removed from the system at will. Specifically, the sampled data are generated under either H_1 or H_0; thus, such data cannot be used

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to test both H_1 and H_0 for a non-separable system. Furthermore, based on the DC framework and the advantages of TE and EC, we proposed a criterion called EE to detect DC between variables. Together with its conditional version, EE and cEE can solve both nonlinear and non-separable problems of causal detection and reduce the scale bias. Thus, EE and cEE are promising indices to quantify total and conditional/direct DC, respectively.

There are still several issues that require discussion and further work:

(1) We only discussed global DC with an autonomous system, but concepts of time-dependent DC and local DC can be proposed in a similar manner. For a time-varying system, time-dependent DC for a short period of time can be also calculated by EE/cEE with sufficient samples. Though the EE/cEE indices can be used, as the cause and effect relationship between variables changes with time, the theory needs to be generalized.
(2) A suitable τ in the DC framework should be carefully chosen in practice to represent the time range of the causal influence under consideration. It is usually determined by the specific problem, while the first minimum point of autocorrelation is a simple choice, as is used for GC/TE/EC.
(3) If we infer the causality lag θ_k in equation (2.1) by EE/cEE, we can choose different components of Y^* in equations (2.7) and (2.8) to measure the causality. Comparing different information from X^{NN} to the components of Y^* is one possible way to detect causality lags.
(4) When constructing a causal network, an appropriate threshold value is an important issue. Siggiridou *et al.* [22] suggest several ways for how to choose the threshold, such as by the surrogate data, the sparsity of the network, or the empirical decision. Although surrogate data method is a good way to show the difference of groups for one measure, it is not a suitable approach to evaluate or compare the performances among several indices. Surrogate data are approximation to the H_0 data, but some indices, such as cEC in figure 3, are far from 0 even under the true H_0 data. As the purpose of using the threshold is to distinguish H_0 and H_1, similar to the significance level, 0.01, 0.025 and 0.05 are considered to be proper threshold values for real systems. Besides, electronic supplementary material, figure S2, shows that EE/cEE have the best performances among the indices when the threshold value is in [0.01, 0.04]. However, the choice of threshold value still needs more research and discussion.
(5) Besides the threshold issue, there are various orders to measure the edges in the network and different choices of variables to be conditioned on. Spirtes & Glymour [43] state the celebrated Peter–Clark algorithm. This is an extensive and profound scientific field for network inference, beyond the DC framework and EE/cEE indices we proposed in this study.
(6) In practice, we found that EE/cEE are very efficient and applicable to chaotic systems with non-zero inner dimension; however, they are less efficient when applied to a system at a stable fixed point with zero inner dimension, where noise dominates the dynamics. This could be an advantage of prior indices (GC and TE) in the original state space over the newer indices (EC and EE) in the delay embedding space. Besides, EC and EE for real systems also rely on the fact that the attractive manifold should be low-dimensional, which makes the indices computable from data. Any solvable high-dimensional problem is in nature a low-dimensional one.
(7) Our DC framework extremely relies on the temporal information contained in the dynamical time series. However, the statistical models [5,6,10], as we mentioned in the introduction section, try to detect causalities for random variables even without time labels. They aim to use the least model constraints to give a weak definition of causality, while DC holds the belief that only temporal flow leads to causality.
(8) Moreover, Pearl's do-calculus [10] considers the interventional causality. Pearl proposed the back-door and front-door criteria to show a way to estimate the interventional causality only by the observed data, which means we can get $P(Y \mid do(X))$ from $P(X, Y, Z_1, Z_2, \ldots)$ and a directed acyclic graph (DAG) even if there are hidden variables. However, the flaw is that the criteria are limited on DAGs. There are usually bidirectional edges (feedback) for real dynamical systems. The combination of DC and do-calculus to estimate the interventional causality only by the observed data is a promising future topic.
(9) Applying causal inferring models and indices to real-world data is one of the purposes of theoretical researches. We used three real examples in this study to show the performance of EE/cEE. However, models and methods need to be improved step by step in wide-ranging real applications. There are many issues affecting the effectiveness of causal models, such as strong noise of the observed data and weak causal association, which should be considered as future topics.

Data accessibility. The data are provided in electronic supplementary material [44]. The codes for calculating embedding entropy/conditional embedding entropy are available at https://github.com/smsxiaomayi/DCcode/blob/main/DCcode.zip. The way to generate simulated datasets and the sources of real datasets are described in detail in the electronic supplementary material.

Authors' contributions. J.S.: conceptualization, data curation, formal analysis, investigation, methodology, resources, software, validation, visualization, writing—original draft, writing—review and editing; C.L.: conceptualization, data curation, funding acquisition, methodology, project administration, supervision, writing—original draft, writing—review and editing; K.A.: funding acquisition, project administration, supervision, writing—original draft, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Competing interests. We declare we have no competing interests.

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References

1. Falcon A. 2019 Aristotle on causality. In *The Stanford encyclopedia of philosophy* (ed. EN Zalta), Spring 2019 edn. Stanford, CA: Metaphysics Research Lab, Stanford University.
2. Lewis D. 1979 Counterfactual dependence and time's arrow. *Nous* **13**, 455–476. (doi:10.2307/2215339)
3. Simon HA, Rescher N. 1966 Cause and counterfactual. *Phil. Sci.* **33**, 323–340. (doi:10.1086/288105)
4. Rubin DB. 2005 Causal inference using potential outcomes: design, modeling, decisions. *J. Am. Stat. Assoc.* **100**, 322–331. (doi:10.1198/016214504000001880)
5. Neyman JS. 1990 On the application of probability theory to agricultural experiments. Essay on principles. Section 9. *Stat. Sci.* **5**, 465–472. (doi:10.1214/ss/1177012031)
6. Rubin DB. 1978 Bayesian inference for causal effects: the role of randomization. *Ann. Stat.* **6**, 34–58. (doi:10.1214/aos/1176344064)
7. Sekhon JS. 2008 The Neyman-Rubin model of causal inference and estimation via matching methods. In *The Oxford handbook of political methodology*. Oxford, UK: Oxford University Press.
8. Pearl J. 1995 Causal diagrams for empirical research. *Biometrika* **82**, 669–688. (doi:10.1093/biomet/82.4.669)
9. Pearl J. 2009 Causal inference in statistics: an overview. *Stat. Surv.* **3**, 96–146. (doi:10.1214/09-SS057)
10. Pearl J. 2000 *Causality: models, reasoning, and inference*. Cambridge, UK: Cambridge University Press.
11. Granger CWJ. 1969 Investigating causal relations by econometric models and cross-spectral methods. *Econometrica* **37**, 424–438. (doi:10.2307/1912791)
12. Schreiber T. 2000 Measuring information transfer. *Phys. Rev. Lett.* **85**, 461–464. (doi:10.1103/PhysRevLett.85.461)
13. Bossomaier T, Barnett L, Harré M, Lizier JT. 2016 *An introduction to transfer entropy*, pp. 65–95. Berlin, Germany: Springer.
14. Hirata Y, Aihara K. 2010 Identifying hidden common causes from bivariate time series: a method using recurrence plots. *Phys. Rev. E* **81**, 016203. (doi:10.1103/PhysRevE.81.016203)
15. Sugihara G, May R, Ye H, Hsieh C-H, Deyle E, Fogarty M, Munch S. 2012 Detecting causality in complex ecosystems. *Science* **338**, 496–500. (doi:10.1126/science.1227079)
16. Lusch B, Maia PD, Kutz JN. 2016 Inferring connectivity in networked dynamical systems: challenges using Granger causality. *Phys. Rev. E* **94**, 032220. (doi:10.1103/PhysRevE.94.032220)
17. Mønster D, Fusaroli R, Tylén K, Roepstorff A, Sherson JF. 2017 Causal inference from noisy time-series data: testing the convergent cross-mapping algorithm in the presence of noise and external influence. *Future Gener. Comput. Syst.* **73**, 52–62. (doi:10.1016/j.future.2016.12.009)
18. Stepaniants G, Brunton BW, Kutz JN. 2020 Inferring causal networks of dynamical systems through transient dynamics and perturbation. *Phys. Rev. E* **102**, 042309. (doi:10.1103/PhysRevE.102.042309)
19. Yuan AE, Shou W. 2021 Data-driven causal analysis of observational time series in ecology. *bioRxiv*. (doi:10.1101/2020.08.03.233692)
20. Paluš M, Krakovská A, Jakubík J, Chvosteková M. 2018 Causality, dynamical systems and the arrow of time. *Chaos* **28**, 075307. (doi:10.1063/1.5019944)
21. Runge J. 2018 Causal network reconstruction from time series: from theoretical assumptions to practical estimation. *Chaos* **28**, 075310. (doi:10.1063/1.5025050)
22. Siggiridou E, Koutlis C, Tsimpiris A, Kugiumtzis D. 2019 Evaluation of Granger causality measures for constructing networks from multivariate time series. *Entropy* **21**, 1080. (doi:10.3390/e21111080)
23. Papana A, Kyrtsou C, Kugiumtzis D, Diks C. 2013 Simulation study of direct causality measures in multivariate time series. *Entropy* **15**, 2635–2661. (doi:10.3390/e15072635)
24. Bakhshayesh H, Fitzgibbon SP, Janani AS, Grummett TS, Pope KJ. 2019 Detecting connectivity in EEG: a comparative study of data-driven effective connectivity measures. *Comput. Biol. Med.* **111**, 103329. (doi:10.1016/j.compbiomed.2019.103329)
25. He Y-B, Geng Z, Liang X. 2005 Learning causal structures based on Markov equivalence class. In *Algorithmic learning theory, 8–11 Octorber 2005, Singapore* (eds S Jain, HU Simon, E Tomita), pp. 92–106. Berlin, Germany: Springer. (doi:10.1007/11564089_9)
26. Leng S, Ma H, Kurths J, Lai Y-C, Lin W, Aihara K, Chen L. 2020 Partial cross mapping eliminates indirect causal influences. *Nat. Commun.* **11**, 2632. (doi:10.1038/s41467-020-16238-0)
27. Rulkov NF, Sushchik MM, Tsimring LS, Abarbanel HD. 1995 Generalized synchronization of chaos in directionally coupled chaotic systems. *Phys. Rev. E* **51**, 980–994. (doi:10.1103/PhysRevE.51.980)
28. Ye H, Deyle ER, Gilarranz LJ, Sugihara G. 2015 Distinguishing time-delayed causal interactions using convergent cross mapping. *Sci. Rep.* **5**, 14750. (doi:10.1038/srep14750)
29. Oizumi M, Tsuchiya N, Amari S-I. 2016 Unified framework for information integration based on information geometry. *Proc. Natl Acad. Sci. USA* **113**, 14 817–14 822. (doi:10.1073/pnas.1603583113)
30. Casdagli M, Sauer T, Yorke JA. 1991 Embedology. *J. Stat. Phys.* **65**, 579–616. (doi:10.1007/BF01053745)
31. Stark J, Broomhead D, Davies M, Huke J. 1997 Takens embedding theorems for forced and stochastic systems. *Nonlinear Anal.: Theory Methods Appl.* **30**, 5303–5314. (doi:10.1016/S0362-546X(96)00149-6)
32. Stark J, Broomhead DS, Davies ME, Huke J. 2003 Delay embeddings for forced systems. II. Stochastic forcing. *J. Nonlinear Sci.* **13**, 519–577. (doi:10.1007/s00332-003-0534-4)
33. Shi J, Zhao J, Liu X, Chen L, Li T. 2018 Quantifying direct dependencies in biological networks by multiscale association analysis. *IEEE/ACM Trans. Comput. Biol. Bioinf.* **17**, 449–458. (doi:10.1109/TCBB.8857)
34. Shi J, Zhao J, Li T, Chen L. 2019 Detecting direct associations in a network by information theoretic approaches. *Sci. China Math.* **62**, 823–838. (doi:10.1007/s11425-017-9206-0)
35. Kraskov A, Stögbauer H, Grassberger P. 2004 Estimating mutual information. *Phys. Rev. E* **69**, 066138. (doi:10.1103/PhysRevE.69.066138)
36. Kozachenko L, Leonenko NN. 1987 Sample estimate of the entropy of a random vector. *Probl. Pered. Inform.* **23**, 9–16.
37. Lindner M, Vicente R, Priesemann V, Wibral M. 2011 Trentool: a Matlab open source toolbox to analyse information flow in time series data with transfer entropy. *BMC Neurosci.* **12**, 119. (doi:10.1186/1471-2202-12-119)
38. Wong TW, Lau TS, Yu TS, Neller A, Wong SL, Tam W, Pang SW. 1999 Air pollution and hospital admissions for respiratory and cardiovascular diseases in Hong Kong. *Occup. Environ. Med.* **56**, 679–683. (doi:10.1136/oem.56.10.679)
39. Mustafić H *et al.* 2012 Main air pollutants and myocardial infarction: a systematic review and meta-analysis. *JAMA* **307**, 713–721. (doi:10.1001/jama.2012.126)
40. Shah AS, Langrish JP, Nair H, McAllister DA, Hunter AL, Donaldson K, Newby DE, Mills NL. 2013 Global association of air pollution and heart failure: a systematic review and meta-analysis. *Lancet* **382**, 1039–1048. (doi:10.1016/S0140-6736(13)60898-3)
41. Benincà E, Jöhnk KD, Heerkloss R, Huisman J. 2009 Coupled predator–prey oscillations in a chaotic food web. *Ecol. Lett.* **12**, 1367–1378. (doi:10.1111/j.1461-0248.2009.01391.x)
42. Benincà E, Huisman J, Heerkloss R, Jöhnk KD, Branco P, Van Nes EH, Scheffer M, Ellner SP. 2008 Chaos in a long-term experiment with a plankton community. *Nature* **451**, 822–825. (doi:10.1038/nature06512)
43. Spirtes P, Glymour C. 1991 An algorithm for fast recovery of sparse causal graphs. *Soc. Sci. Comput. Rev.* **9**, 62–72. (doi:10.1177/089443939100900106)
44. Shi J, Chen L, Aihara K. 2022 Supplementary Information for "Embedding entropy: a nonlinear measure of dynamical causality". Figshare.

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